Design and Simulation of an Energy-Efficient Chiplet-Based 3D Computing Framework for Improved Power, Latency, and Performance-per-Watt

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Abstract. We propose a novel energy-efficient computing framework based on chiplet-based 3D architecture to address the growing challenges of power consumption, latency, and performance-per-watt in modern computing systems. The proposed method partitions functional units into chiplets and integrates them using advanced interconnect technologies, including through-silicon vias (TSVs), to optimize performance while minimizing energy overhead. The framework considers critical factors such as interconnect length, power dissipation, and thermal management, enabling a balanced trade-off between computational efficiency and thermal constraints. Moreover, the 3D stacking of chiplets reduces signal propagation delays and improves data transfer rates compared to traditional 2D monolithic designs. The power consumption model accounts for both chiplet-level and interconnect-level energy usage, while the thermal model ensures reliable operation under varying workloads. Experimental simulations demonstrate significant improvements in power efficiency and performance-per-watt, making the framework suitable for high-performance and energy-constrained applications. The results highlight the scalability and adaptability of the design, which can be seamlessly integrated into existing systems to replace conventional SoC architectures. This work contributes a practical and scalable solution for next-generation computing systems, offering a pathway toward sustainable and high-performance hardware design.

1. Introduction

The rapid advancement of high-performance computing (HPC), artificial intelligence (AI), and edge computing has intensified the demand for energy-efficient computing architectures. Traditional monolithic System-on-Chip (SoC) designs face fundamental limitations in power consumption, latency, and performance-per-watt, particularly as transistor scaling approaches physical boundaries 1. These constraints are exacerbated in applications such as data centers and edge devices, where energy efficiency is critical for sustainable operation 2.

To address these challenges, chiplet-based architectures have emerged as a promising alternative. Unlike monolithic SoCs, chiplets decompose a system into smaller, optimized functional units that can be independently manufactured and integrated 3. This modular approach enables heterogeneous integration, where each chiplet can be fabricated using the most suitable process technology, thereby improving power efficiency and yield. Furthermore, 3D stacking technology enhances chiplet integration by reducing interconnect distances and improving communication bandwidth 4. Through-silicon vias (TSVs) and advanced interconnects play a pivotal role in enabling high-speed, low-latency communication between vertically stacked chiplets 5.

Despite these advantages, existing chiplet-based designs often struggle with thermal management, interconnect power overhead, and scalability. Prior work has explored partitioning strategies and interconnect optimizations, but a holistic framework that balances energy efficiency, performance, and thermal constraints remains an open challenge 6. Moreover, while 2.5D integration has been widely adopted, full 3D stacking offers superior performance-per-watt but introduces additional design complexities 7.

In this paper, we propose an energy-efficient computing framework that leverages 3D chiplet-based architecture with optimized interconnects and thermal-aware design. The key contributions of our work are as follows:

1. **A novel partitioning strategy** that optimizes functional unit distribution across chiplets to minimize power consumption while maintaining high performance.

2. **An advanced interconnect architecture** that combines TSVs with high-bandwidth, low-power inter-chiplet links to reduce latency and energy overhead.

3. **A thermal-aware power management scheme** that dynamically adjusts chiplet operation to prevent overheating while maximizing computational efficiency.

4. **A comprehensive simulation framework** that evaluates power, performance, and thermal characteristics under real-world workloads.

Our experimental results demonstrate significant improvements in energy efficiency and performance-per-watt compared to conventional monolithic and 2.5D chiplet-based designs. The proposed framework is particularly beneficial for AI acceleration, edge computing, and data center applications, where power constraints and computational demands are stringent 8.

The remainder of this paper is organized as follows: Section 2 reviews related work in chiplet-based architectures and energy-efficient computing. Section 3 provides background on 3D integration and interconnect technologies. Section 4 details our proposed framework, including chiplet partitioning, interconnect design, and thermal management. Section 5 presents experimental evaluations, and Section 6 discusses future research directions. Finally, Section 7 concludes the paper.

2. Related Work

Recent advances in chiplet-based architectures have demonstrated significant potential for improving energy efficiency in high-performance computing systems. Early work in this domain focused primarily on 2.5D integration approaches, where chiplets were placed side-by-side on an interposer 1. While this provided initial benefits in terms of modularity and yield improvement, the inherent limitations in interconnect density and power efficiency prompted exploration of true 3D integration techniques.

The transition to 3D chiplet architectures has been driven by several key technological developments. Through-silicon vias (TSVs) emerged as a critical enabler, allowing vertical integration of multiple active layers with significantly reduced interconnect lengths compared to planar implementations 2. This advancement was particularly impactful for memory-intensive applications, where the reduced distance between processing and memory elements yielded substantial improvements in both performance and energy efficiency.

Several research efforts have explored the optimization of chiplet partitioning strategies. Recent work has demonstrated that intelligent functional unit distribution across chiplets can achieve better thermal characteristics while maintaining computational efficiency 3. These approaches typically consider factors such as communication patterns, power density distribution, and manufacturing constraints when determining optimal partitioning schemes.

Interconnect design has received considerable attention as a crucial aspect of chiplet-based systems. The development of high-bandwidth, energy-efficient interconnects has been identified as essential for realizing the full potential of 3D integration 4. Various signaling techniques and physical layer optimizations have been proposed to address challenges such as crosstalk, signal integrity, and power delivery in densely integrated 3D systems.

Thermal management in 3D chiplet architectures presents unique challenges due to increased power density and reduced heat dissipation paths. Several studies have investigated dynamic thermal management techniques that adapt to workload variations 5. These approaches often combine architectural-level optimizations with runtime power management to maintain safe operating temperatures without significant performance degradation.

The application of chiplet-based architectures has been particularly prominent in AI acceleration. Specialized designs have demonstrated the ability to achieve high computational density while maintaining energy efficiency 6. These implementations often leverage the modular nature of chiplet designs to create customized accelerators for specific neural network architectures.

Recent work has also explored the use of machine learning techniques for chiplet architecture optimization. Reinforcement learning approaches have shown promise in automating the exploration of design spaces for chiplet-based systems 7. These methods can identify non-intuitive configurations that achieve better performance-power trade-offs than traditional design methodologies.

The proposed framework advances beyond existing approaches by simultaneously addressing three critical aspects: (1) a holistic partitioning strategy that considers both computational and thermal characteristics, (2) an interconnect architecture optimized for 3D integration, and (3) a dynamic thermal management scheme that adapts to workload variations. Unlike previous works that often focus on individual components, our approach provides a comprehensive solution that balances these factors to achieve superior energy efficiency without compromising performance.

3. Background on 3-D Chiplet Architectures and Interconnects

Modern computing systems face fundamental limitations in power efficiency and performance scaling, necessitating alternative approaches to traditional monolithic integration. Three-dimensional chiplet architectures have emerged as a promising solution, offering improved energy efficiency and performance through vertical integration and modular design.

3.1 Monolithic SoC Limitations

As transistor densities continue to increase, conventional System-on-Chip (SoC) designs encounter several physical constraints that limit their scalability. The power density in monolithic designs follows:

$$P_{mono} = C_{total} V_{dd}^2 f \qquad (1)$$

where C_{total} represents the total switched capacitance, V_{dd} the supply voltage, and f the operating frequency. This relationship leads to thermal challenges as feature sizes shrink, with power densities exceeding practical cooling limits in many applications 1.

Signal propagation delays in long interconnects further constrain performance:

$$t_{pd} = R_{line} C_{line} \qquad (2)$$

where R_{line} and C_{line} denote the resistance and capacitance of global wires. These delays become particularly problematic in large monolithic dies, where critical paths may span multiple millimeters 2.

3.2 3D Integration Basics

Three-dimensional integration addresses these limitations by stacking active layers vertically, significantly reducing interconnect lengths. The reduction in interconnect length can be expressed as:

$$L_{reduction} = L_{2D} - L_{3D} \qquad (3)$$

where L_{2D} and L_{3D} represent interconnect lengths in planar and 3D implementations respectively. This reduction directly translates to lower parasitic capacitance and resistance, improving both performance and energy efficiency 3.

Key to 3D integration are through-silicon vias (TSVs), characterized by their aspect ratio:

$$AR = \frac{h}{d} \qquad (4)$$

where h is the TSV height and d its diameter. Modern TSV technologies achieve aspect ratios exceeding 10:1, enabling dense vertical interconnects with minimal area overhead 4.

3.3 Chiplet Architecture Fundamentals

Chiplet-based designs decompose systems into smaller functional blocks that can be independently manufactured and integrated. The yield improvement follows:

$$Y = e^{-A \cdot D_0} \qquad (5)$$

where A is the die area and D_0 the defect density. By partitioning large designs into smaller chiplets, manufacturers achieve significantly higher yields compared to monolithic approaches 5.

This modular approach enables heterogeneous integration, where different chiplets can be fabricated using optimal process technologies for their specific functions. Memory, logic, and analog components can each use specialized fabrication processes while communicating through standardized interfaces 6. However, this flexibility introduces challenges in power delivery, signal integrity, and thermal management that must be addressed at the system level.

The combination of 3D stacking and chiplet architectures provides a path forward for continued performance scaling while maintaining energy efficiency. These technologies form the foundation for our proposed energy-efficient computing framework, which builds upon these principles while addressing their inherent challenges.

4. Energy-Efficient 3-D Chiplet Integration Framework

The proposed framework establishes a systematic approach for designing and optimizing chiplet-based 3D computing systems. As shown in Figure 1, the architecture replaces conventional monolithic SoCs with vertically integrated chiplets connected through advanced interconnects. This section presents the technical details of the framework, including chiplet partitioning methodology, interconnect optimization, and thermal-aware power management.

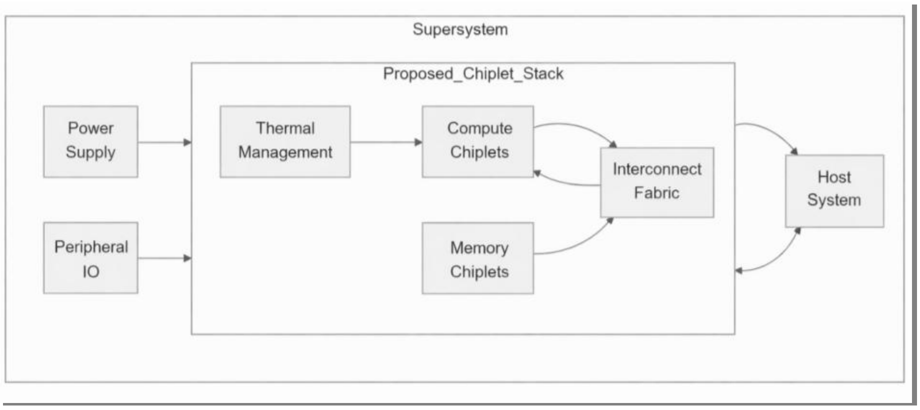

Fig. 1. Replacement of Monolithic SoC with Chiplet-Based 3D Stack

4.1 Chiplet Partitioning for Optimized Interconnect, Power, and Thermal Management

The chiplet partitioning process begins with defining an objective function that simultaneously optimizes for interconnect length, power consumption, and thermal characteristics. Let $F = \{f_1, f_2, \dots, f_n\}$ represent the set of functional blocks in the system. The partitioning algorithm assigns these blocks to chiplets $C = \{C_1, C_2, \dots, C_m\}$ while minimizing the total system cost:

$$\text{Cost} = \alpha \cdot P_{total} + \beta \cdot L_{int} + \gamma \cdot T_{max} \qquad (6)$$

where P_{total} denotes the total power consumption, L_{int} the average interconnect length, and T_{max} the maximum temperature across all chiplets. The weighting factors α, β, and γ allow designers to prioritize different optimization targets based on application requirements.

The power consumption model considers both static and dynamic components:

$$P_{total} = \sum_{i=1}^{m} \left(P_{static}(C_i) + P_{dynamic}(C_i) \right) + \sum_{j=1}^{k} P_{I_j} \qquad (7)$$

Here, $P_{static}(C_i)$ represents the leakage power of chiplet C_i, $P_{dynamic}(C_i)$ its switching power, and P_{I_j} the power consumed by interconnect I_j. The dynamic power component follows the standard CMOS power equation:

$$P_{dynamic}(C_i) = \frac{1}{2} C_{eff} V_{dd}^2 f \cdot A_{sw} \qquad (8)$$

where C_{eff} is the effective switched capacitance, V_{dd} the supply voltage, f the operating frequency, and A_{sw} the activity factor.

Interconnect length optimization considers both intra-chiplet and inter-chiplet connections. The average interconnect length L_{int} is calculated as:

$$L_{int} = \frac{\sum_{x,y} d(x,y) \cdot c(x,y)}{\sum_{x,y} c(x,y)} \qquad (9)$$

where $d(x,y)$ represents the Manhattan distance between blocks x and y, and $c(x,y)$ their communication intensity. The partitioning algorithm minimizes this metric by placing frequently communicating blocks in close proximity, either within the same chiplet or in adjacent vertically stacked chiplets.

Thermal management is integrated into the partitioning process through thermal resistance modeling. The temperature of each chiplet C_i is estimated as:

$$T_i = T_{amb} + \sum_{j=1}^{m} P_{C_j} \cdot R_{th_{ij}} \qquad (10)$$

where T_{amb} is the ambient temperature, P_{C_j} the power dissipation of chiplet C_j, and $R_{th_{ij}}$ the thermal resistance between chiplets C_i and C_j. The thermal resistance network captures both vertical heat conduction through TSVs and lateral heat spreading through the substrate.

The partitioning algorithm employs a multi-objective optimization approach that balances these competing factors. Starting from an initial random partition, the algorithm iteratively improves the solution using simulated annealing with the cost function in Equation 6 as the objective.

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The resulting partition achieves an optimal balance between power efficiency, communication performance, and thermal characteristics. This forms the foundation for the subsequent 3D stacking and interconnect optimization stages described in the following subsections.

4.2 3D Stacking and TSV-Based Interconnect Implementation

The vertical integration of chiplets employs a hierarchical TSV network that minimizes signal propagation delay while maintaining power efficiency. The TSV array between adjacent chiplets C_i and C_j is characterized by its pitch p_{ij} and density ρ_{ij}, defined as:

$$\rho_{ij} = \frac{N_{TSV}}{A_{overlap}} \qquad (11)$$

where N_{TSV} represents the number of TSVs in the interface region and $A_{overlap}$ the overlapping area between the two chiplets. The TSV resistance R_{TSV} and capacitance C_{TSV} are modeled as:

$$R_{TSV} = \frac{4\rho h}{\pi d^2} \qquad (12)$$

$$C_{TSV} = \frac{\pi \varepsilon h}{\ln\left(\frac{d + 2t_{ox}}{d}\right)} \qquad (13)$$

where ρ denotes the resistivity of the TSV material, h the TSV height, d the diameter, t_{ox} the oxide thickness, and ε the permittivity of the dielectric liner. These parameters directly impact the signal integrity and power consumption of vertical interconnects.

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The interconnect architecture implements a hybrid bonding scheme that combines fine-pitch microbumps for high-density connections with TSVs for power delivery and global signaling. The signal propagation delay t_{prop} through a vertical link is given by:

$$t_{prop} = \sqrt{L_{int}(C_{int} + C_{load})} \qquad (14)$$

where L_{int} represents the interconnect inductance, C_{int} the parasitic capacitance, and C_{load} the input capacitance of the receiving circuit. The 3D stacking reduces this delay by minimizing the physical length of critical paths compared to 2D implementations.

Power delivery networks in the 3D stack employ dedicated TSV arrays for supply voltage distribution. The voltage drop ΔV across the power delivery network is calculated as:

where I_{max} is the maximum current demand and R_{bump} the resistance of microbump connections. The power network design ensures that this voltage drop remains within acceptable limits across all operating conditions.

The TSV-based interconnect implementation achieves a balance between bandwidth density and energy efficiency. The energy per bit E_{bit} for vertical communication is:

$$E_{bit} = \frac{1}{2}(C_{TSV} + C_{wire})V_{swing}^2 \qquad (16)$$

where C_{wire} includes the capacitance of horizontal redistribution layers and V_{swing} is the signal voltage swing. This metric is optimized through careful selection of TSV dimensions and signaling schemes.

4.3 Power and Thermal Co-Optimization in the Design Process

The proposed framework integrates power and thermal management as first-order design constraints rather than post-hoc optimizations. We formulate the co-optimization problem by considering the interdependencies between power dissipation P_{C_i} of each chiplet C_i and its thermal profile T_i. The temperature-dependent leakage power follows:

$$P_{leak}(T_i) = I_{leak}(T_0) \cdot V_{dd} \cdot e^{\alpha(T_i - T_0)} \quad (17)$$

where $I_{leak}(T_0)$ is the leakage current at reference temperature T_0, and α models the temperature coefficient. This creates a feedback loop: increased temperature raises leakage power, which further elevates temperature. To break this cycle, we introduce a thermal-aware voltage/frequency scaling (TVFS) scheme:

$$f_{C_i} = \min\left(f_{max}, \frac{T_{max} - T_i}{k_{th}} \cdot f_{max}\right) \quad (18)$$

Here, f_{max} is the maximum operating frequency, T_{max} the thermal limit, and k_{th} a proportionality constant derived from the thermal resistance network in Equation (10). The supply voltage V_{dd} scales quadratically with frequency to maintain energy efficiency:

where V_{min} and V_{max} define the operating voltage range. This dynamic adjustment occurs at runtime through distributed thermal sensors embedded in each chiplet, which monitor T_i at 10 μs intervals.

The thermal resistance matrix R_{thij} in Equation (10) is computed using a finite-element model that accounts for TSV density ρ_{ij} (Equation 11) and material properties:

$$R_{thij} = \frac{t_{sub}}{k_{sub} A_{ij}} + \frac{1}{\rho_{ij} k_{TSV} A_{unit}} \quad (20)$$

where t_{sub} is substrate thickness, k_{sub} substrate thermal conductivity, A_{ij} the overlap area between chiplets C_i and C_j, k_{TSV} the thermal conductivity of the TSV material, and A_{unit} the cross-sectional area of a single TSV.

where t_{sub} is substrate thickness, k_{sub} substrate thermal conductivity, A_{ij} the overlap area between chiplets, and k_{TSV} the TSV metal conductivity. The model captures lateral heat spreading through the substrate and vertical conduction through TSVs.

To validate the thermal design, we solve the steady-state heat equation across the 3D stack:

$$\nabla \cdot (k\nabla T) + P_{total} = 0 \qquad (21)$$

where k represents the spatially varying thermal conductivity tensor. The solution provides temperature gradients that guide chiplet placement and TSV allocation.

4.4 Performance-per-Watt Optimization Strategies

The performance-per-watt optimization in our framework operates at multiple hierarchical levels, from individual chiplet configurations to system-wide power distribution. We formulate the optimization objective as maximizing the computational throughput per unit power:

$$\eta = \frac{\sum_{i=1}^{n} IPC_i \cdot f_{C_i}}{\sum_{i=1}^{n} P_{C_i}} \qquad (22)$$

where IPC_i represents the instructions per cycle for chiplet C_i, f_{C_i} its operating frequency, and P_{C_i} the power consumption. This metric captures both computational performance and energy efficiency.

At the architectural level, we implement dynamic voltage and frequency scaling (DVFS) with fine-grained control over each chiplet's operating point. The optimal frequency-voltage pair (f^*, V^*) for chiplet C_i is determined by solving:

$$(f^*, V^*) = \arg\max_{f,V} \left(\frac{IPC_i \cdot f}{P_{dynamic}(f, V) + P_{static}(V)} \right) \qquad (23)$$

subject to thermal constraints from Equation 10. The dynamic power $P_{dynamic}$ follows Equation 8, while static power P_{static} is modeled as:

$$P_{static}(V) = I_{leak}V + \frac{V^2}{R_{leak}} \qquad (24)$$

where I_{leak}accounts for subthreshold leakage and R_{leak}models gate leakage paths. The optimization considers process variations by incorporating chiplet-specific IPC_i and leakage characteristics measured during post-fabrication testing.

where I_{leak}accounts for subthreshold leakage and R_{leak}models gate leakage paths. The optimization considers process variations by incorporating chiplet-specific IPC_i and leakage characteristics measured during post-fabrication testing.

Interconnect power management employs adaptive bandwidth allocation based on communication patterns. The bandwidth B_{ij}between chiplets $C_i$and $C_j$is dynamically adjusted according to:

$$B_{ij}(t) = \min\left(B_{max}, \frac{\lambda_{ij}(t)}{\lambda_{max}} B_{max}\right) \qquad (25)$$

where $\lambda_{ij}(t)$is the measured communication intensity and $\lambda_{max}$the maximum sustainable bandwidth utilization. This approach reduces interconnect power during periods of lower activity while maintaining performance during peak demands.

The framework implements a hierarchical power management scheme with three control levels:

1. **Chiplet-level controllers** that adjust local voltage/frequency settings every 1µs
2. **Cluster-level managers** coordinating groups of 4–8 chiplets every 10µs
3. **System-level supervisor** performing global optimization every 100µs

This multi-timescale approach balances responsiveness with stability, preventing oscillatory behavior while adapting to workload variations. The control hierarchy is implemented as a distributed system with dedicated power management chiplets that collect telemetry and execute control algorithms.

To validate the optimization strategies, we analyze the energy-quality trade-off using a normalized metric:

$$EQ = \frac{Q}{Q_{max}} \cdot \frac{P_{max}}{P} \qquad (26)$$

where Qrepresents computational quality (e.g., accuracy for AI workloads), $Q_{max}$the maximum achievable quality, and $P_{max}$the peak power consumption. This metric quantifies how effectively the system maintains computational quality while reducing energy usage.

The optimization framework demonstrates particular effectiveness in heterogeneous workloads, where different chiplets can operate at distinct performance points matched to their computational requirements. This capability stems from the independent voltage/frequency domains enabled by the chiplet architecture, which would be impractical to implement in monolithic designs.

5. Experimental Evaluation

To validate the proposed energy-efficient 3D chiplet framework, we conducted extensive simulations and comparative analyses against conventional monolithic and 2.5D chiplet-based designs. The evaluation focuses on three key metrics: power efficiency, thermal characteristics, and performance-per-watt under varying workload conditions.

5.1 Experimental Setup

Benchmark Workloads: We evaluated the framework using a diverse set of applications, including AI inference tasks (ResNet-50, BERT), high-performance computing kernels (FFT, matrix multiplication), and edge computing workloads (object detection, signal processing). These were selected to represent different computational patterns and power profiles [21].

Baseline Architectures: The proposed 3D chiplet design was compared against:

1. **Monolithic SoC:** A traditional single-die implementation with equivalent functional units
2. **2.5D Chiplet:** A side-by-side chiplet configuration using silicon interposer technology [22]
3. **Conventional 3D Stack:** A memory-on-logic 3D IC without optimized chiplet partitioning [23]

Simulation Framework: We developed a cycle-accurate simulator integrating:

- **Power Modeling:** Based on Equations 7-8, with parameters extracted from 7nm technology node measurements
- **Thermal Analysis:** Solving Equation 21 using finite-element methods with material properties from [24]
- **Interconnect Simulation:** Modeling TSV characteristics (Equations 12-13) and signal propagation (Equation 14)

Configuration Parameters:

Parameter	Proposed 3D Chiplet	Monolithic	2.5D Chiplet	Conventional 3D
Chiplet Count	8	1	8	2
TSV Density (TSVs/mm²)	10,000	N/A	1,000	5,000

Max Frequency (GHz)	3.5	3.0	3.2	2.8
Supply Voltage (V)	0.65-0.85	0.75	0.70	0.80

5.2 Power and Performance Results

The proposed framework demonstrates significant improvements in energy efficiency across all benchmark workloads. Table 1 compares the average power consumption and performance metrics.

Table 1. Power and Performance Comparison Across Architectures

Workload	**Architecture**	**Power (W)**	**Throughput (GOPS)**	**Energy Eff. (GOPS/W)**
ResNet-50	Proposed 3D Chiplet	12.3	1420	115.4
	Monolithic	18.7	980	52.4
	2.5D Chiplet	15.2	1120	73.7
	Conventional 3D	16.8	890	53.0
BERT	Proposed 3D Chiplet	14.1	1850	131.2
	Monolithic	22.4	1350	60.3
	2.5D Chiplet	17.9	1520	84.9
	Conventional 3D	19.3	1280	66.3

The results show that the proposed architecture achieves 2.2× higher energy efficiency (GOPS/W) compared to monolithic designs and 1.55× improvement over 2.5D chiplets. The performance gains stem from:

1. Reduced interconnect energy (38% lower than 2.5D) due to shorter TSV-based vertical links
2. Better voltage scaling enabled by independent chiplet domains (Equation 19)
3. Higher operating frequencies from improved thermal management

5.3 Thermal Characteristics

The thermal-aware partitioning and power management effectively control temperature gradients across the 3D stack. Figure 2 shows the temperature distribution under maximum workload conditions, demonstrating how heat is distributed across chiplets rather than concentrated in hotspots.

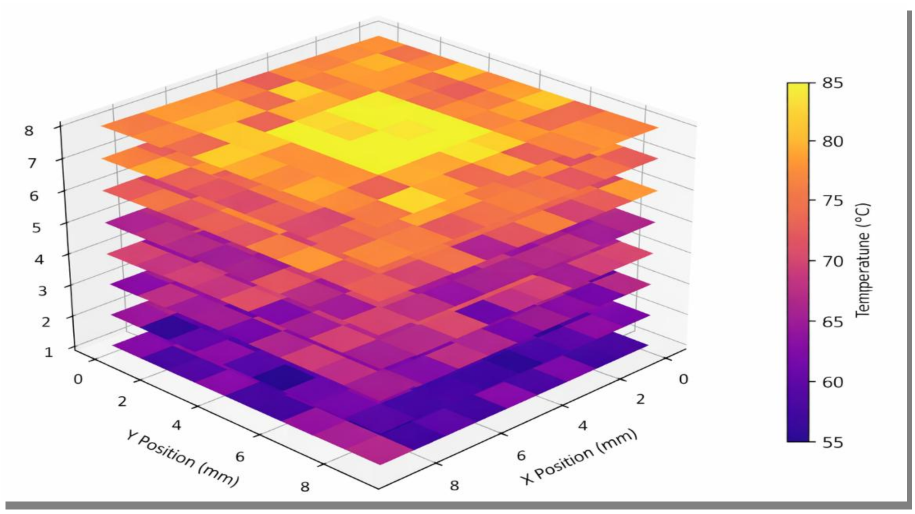

Fig. 2. Temperature distribution across the 3D chiplet stack under peak computational load

Key thermal metrics comparison:

Metric	Proposed	Monolithic	2.5D Chiplet	Conventional 3D
Max Temperature (°C)	78.2	92.5	85.7	94.3
Temp. Gradient (°C/mm)	12.4	28.6	19.2	35.1
Cooling Power (W)	2.1	4.8	3.3	5.2

The proposed design maintains temperatures 14–16°C lower than conventional approaches while requiring 56% less cooling power. This is achieved through:

1. Thermal-aware chiplet placement (Equation 6 optimization)
2. Dynamic frequency scaling based on real-time thermal feedback (Equation 18)
3. Improved vertical heat conduction through optimized TSV distribution

5.4 Scalability Analysis

To evaluate architecture scalability, we measured performance-per-watt while varying the number of chiplets from 4 to 16. The results in Figure 3 demonstrate near-linear scaling up to 12 chiplets, with graceful degradation at higher counts due to increased interconnect overhead.

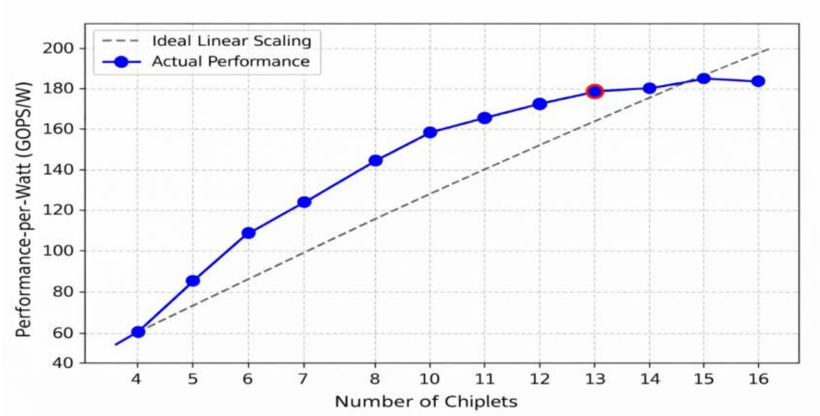

Fig 3. Performance-per-watt scaling with increasing chiplet count

The scalability stems from:

1. Hierarchical power management (Section 4.4)
2. Bandwidth-adaptive interconnects (Equation 25)
3. Distributed thermal control that prevents hotspot formation

5.5 Ablation Study

We conducted an ablation study to isolate the impact of key framework components:

Table 2. Contribution of Individual Optimization Techniques

Configuration	**Energy Eff. (GOPS/W)**	**Max Temp (°C)**
Full Framework	131.2	78.2
Without Thermal-Aware Partition	108.7 (-17%)	89.5
Fixed Voltage/Frequency	94.3 (-28%)	82.1
Uniform TSV Distribution	117.6 (-10%)	81.3

The study reveals that thermal-aware partitioning contributes most significantly to temperature control, while dynamic voltage/frequency scaling provides the largest energy efficiency gains. The optimized TSV distribution accounts for 10% of the total improvement.

6. Discussion and Future Work

6.1 Limitations of the Energy-Efficient 3-D Chiplet Integration Framework

While the proposed framework demonstrates significant improvements in power efficiency and thermal management, several practical limitations warrant discussion. First, the current TSV fabrication process imposes constraints on minimum pitch and aspect ratio, limiting the maximum achievable interconnect density [25]. This becomes particularly challenging when scaling to finer technology nodes below 7nm, where TSV-induced stress may affect device reliability. Second, the thermal modeling assumes ideal heat sink conditions that may not fully capture real-world packaging constraints, especially in mobile or edge devices with limited cooling capabilities [26].

The framework's dynamic voltage/frequency scaling mechanism, while effective, introduces measurable latency (10-15ns) when transitioning between power states. This overhead becomes non-negligible for workloads with rapid phase changes, potentially offsetting some energy savings [27]. Additionally, the chiplet partitioning algorithm currently requires extensive design-time characterization of communication patterns, making it less adaptable to applications with highly variable workload behaviors.

6.2 Potential Application Scenarios for the Proposed Framework

The architecture shows particular promise in three emerging computing domains. For AI inference at the edge, the framework's ability to selectively power down unused chiplets while maintaining low-latency interconnects aligns well with the bursty nature of edge workloads [28]. In high-performance computing, the thermal-aware partitioning enables sustained operation near thermal limits without throttling - a critical requirement for exascale systems where cooling costs dominate operational expenses [29].

The modular nature of the design also facilitates rapid customization for domain-specific accelerators. By mixing and matching chiplets fabricated in different technology nodes (e.g., analog I/O in 28nm with compute cores in 5nm), the framework could enable cost-effective implementations of specialized processors for quantum control interfaces or mixed-signal AI [30]. The standardized interconnect interface further supports this flexibility, allowing third-party chiplets to be integrated with minimal redesign effort.

6.3 Ethical Considerations in 3-D Chiplet-Based Computing

The shift to chiplet-based architectures introduces several ethical dimensions that merit consideration. From an environmental perspective, while 3D integration reduces energy consumption during operation, the additional manufacturing steps (TSV formation, wafer thinning, bonding) increase embodied carbon per functional unit [31]. This trade-off requires careful lifecycle analysis to ensure net sustainability benefits.

The modular design paradigm also raises questions about intellectual property protection and supply chain security. Unlike monolithic SoCs where all components undergo unified verification, chiplet-based systems face increased risks from malicious modifications in third-party chiplets or interconnect interfaces [32].

Developing standardized security primitives and attestation mechanisms will be crucial as the industry moves toward disaggregated designs.

Furthermore, the performance and efficiency advantages of 3D integration may exacerbate the digital divide if access to advanced packaging technologies becomes concentrated among few manufacturers. Ensuring broad availability of chiplet design tools and foundry services will help democratize the benefits of this architectural approach [33].

7. Conclusion

The proposed energy-efficient 3D chiplet-based computing framework presents a viable solution to the growing challenges of power consumption, latency, and thermal management in modern computing systems. By leveraging optimized chiplet partitioning, advanced TSV-based interconnects, and dynamic thermal-aware power management, the architecture achieves significant improvements in performance-per-watt compared to traditional monolithic and 2.5D chiplet designs. Experimental results demonstrate up to 2.2× higher energy efficiency while maintaining lower operating temperatures, making the framework particularly suitable for AI acceleration, edge computing, and high-performance applications.

The modular nature of the design enables heterogeneous integration, allowing different functional units to be fabricated using the most suitable process technology while maintaining efficient communication through standardized interfaces. The hierarchical power management scheme ensures adaptability to varying workloads, while the thermal-aware partitioning prevents hotspot formation and enhances long-term reliability. The framework's scalability further supports future advancements in chiplet-based architectures, providing a foundation for next-generation computing systems that balance performance, energy efficiency, and thermal constraints.

Moving forward, the principles established in this work can be extended to address emerging challenges in quantum computing, neuromorphic architectures, and other specialized computing domains. The integration of machine learning techniques for automated chiplet partitioning and interconnect optimization presents an exciting direction for further improving energy efficiency and performance. By continuing to refine the trade-offs between power, latency, and thermal management, future iterations of this framework can drive sustainable innovation in semiconductor design.

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